

# Validity Analysis: ARABICMMLU

Target context: Non-Arab Tourists and Expats in Eight Arabic-Speaking Countries (ArabicMMLU Deployment)

Overall risk: **HIGH**

## Deployment Context

**Use case:** Support foreigners visiting Morocco, Egypt, Jordan, Palestine, Lebanon, UAE, Kuwait, or KSA in answering their general questions

**Target population:** Non-Arab tourists or expats visiting Morocco, Egypt, Jordan, Palestine, Lebanon, UAE, Kuwait, or KSA and have general questions about the Arabic language, history or geography of these countries

## Dimension Scores

| Dimension         | Score | Rating           | Confidence |
|-------------------|-------|------------------|------------|
| Input Ontology ✓  | 3     | Moderate gaps    | high       |
| Input Content △   | 2     | Significant gaps | high       |
| Input Form ✓      | 3     | Moderate gaps    | high       |
| Output Ontology △ | 1     | Serious concern  | high       |
| Output Content △  | 2     | Significant gaps | high       |
| Output Form △     | 1     | Serious concern  | high       |
| Average           | 2.0   |                  |            |

△ = highest concern    ✓ = strongest dimension

## Dimension Key

| Abbr. | Dimension       | Definition                                                                                                   |
|-------|-----------------|--------------------------------------------------------------------------------------------------------------|
| IO    | Input Ontology  | Whether the benchmark's test case categories cover the query types expected in deployment.                   |
| IC    | Input Content   | Whether individual datapoint content is culturally and contextually appropriate for the target region.       |
| IF    | Input Form      | Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.             |
| OO    | Output Ontology | Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries. |
| OC    | Output Content  | Whether ground-truth labels align with the judgments of regional stakeholders.                               |
| OF    | Output Form     | Whether the expected output modality matches regional deployment needs and accessibility.                    |

## Overall Summary

ArabicMMLU is a serious ground-up Arabic-language knowledge benchmark whose subject taxonomy and authentic source materials provide reasonable construct validity for evaluating school-curriculum knowledge of Arabic language, history, and geography. For the specific deployment of non-Arab tourists/expats across eight Arabic-speaking countries seeking educational-level knowledge, however, two structural mismatches dominate the assessment: (1) the single-correct-answer MCQ output ontology and form fundamentally cannot evaluate the multi-perspective, explanatory natural-language responses the deployment explicitly requires [Q27, Q88, elicitation A3]; and (2) input content and verification skew heavily Jordanian, with Morocco's school-curriculum content largely absent, Kuwait effectively absent, and no Palestinian annotators despite Palestine

being a top-3 source country [Q31, Q89, WEB-3, DATASET-D4, DATASET-D5]. The 96% answer-key verification ceiling [Q43] likely reflects Jordanian/Egyptian consensus rather than pan-regional validity for contested historical and civic content. Strengths include authentic exam sourcing across eight countries, level-appropriate subject coverage, MSA text-only format matching the deployment modality, and confirmed substantial Palestinian STEM/curriculum content.

| ID    | Evidence                                                                                                                                                                                                                                                |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q27   | "Each question has 2–5 candidate answers, with one correct answer." (p.3)                                                                                                                                                                               |
| Q31   | "The data construction process involved a total of 10 Arabic native speakers from different countries: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 from UAE, and 2 from KSA) and 4 external workers (3 Jordanian and 1 Egyptian)." (p.4) |
| Q43   | "We found that 96% of the questions and answer keys match on average, while the remaining could have incorrect answer keys." (p.4)                                                                                                                      |
| Q88   | "For future work, ArabicMMLU can be extended to include short-answer or essay questions, different modalities (i.e., images, audio, video), larger region coverage, and more questions in professional settings." (p.9)                                 |
| Q89   | "ArabicMMLU does not represent all Arabic countries equally. For example, we have collected over 6K multiple-choice questions from Jordan, while other countries are represented with only 100 questions or, in some cases, not at all." (p.9)          |
| D4    | لُدُع... يندرأل ال يبرع ال شي ج ل تال ي ك ش ت ن م ة ي ب ر ع ال ة و ق ل ... 1952 ماع ندرأل ي ف ة ي ف ال ت ئ ا ة م و ك ح ل و ا ل ي ك ش ت م ت<br>All 5 sampled Civics Middle School items are exclusively about Jordanian political history                |
| D5    | ي ن ط و ل ا ع ا ف د ل ا ب ة ف ل ك م ل ا ة ي م و ك ح ل ا ة ط ل س ل و ا ل د ع ل ا ر ي ز و ن م ... ف ا ن ئ ت س ا ل ا ل ج ا ئ د ت ب ي<br>(Professional) items are from Morocco (Moroccan procedural law)                                                    |
| WEB-3 | Palestine confirmed top-3 source country — yet no Palestinian annotators in pool ( <a href="https://github.com/mbzuai-nlp/ArabicMMLU">https://github.com/mbzuai-nlp/ArabicMMLU</a> )                                                                    |

## Practical Guidance

### What This Benchmark Measures

ArabicMMLU measures a model's ability to select curriculum-designated correct answers on MCQ school-exam questions across 40 subjects spanning STEM, social science, humanities, Arabic language, and other domains, drawn predominantly from Jordan with substantial Egyptian and Palestinian content. The strongest dimension for this deployment is Input Ontology (subject coverage approximately matches the deployment's school-curriculum scope) and Input Form (uniform MSA text matches deployment modality). It can usefully assess whether a system has acquired the factual content of Jordanian-curriculum school knowledge in MSA.

### Construct Depth

The benchmark probes factual recall and basic reasoning at primary, middle, and high school levels (university content only 6% [Q71]), but only at the depth of single-correct-answer MCQ. It does not probe explanatory quality, multi-perspective reasoning, perspective-flagging on contested content, robustness to learner-register Arabic, or appropriate framing for non-Muslim/non-Arab audiences — all of which are central to the deployment requirements. The Output Ontology and Output Form dimensions, scored 1, indicate that even a model achieving the 96% verification ceiling [Q44] would not have been evaluated on the pluralistic, explanatory behaviors the deployment demands.

### What Else You Need

For valid deployment evaluation, substantial supplementation is required: (1) an open-ended Arabic generation evaluation such as AraGen with 3C3H LLM-as-judge [WEB-6] or 'Beyond MCQ' [WEB-5] to address the OO/OF gap; (2) a bespoke contested-content evaluation with Palestinian, Moroccan, and Kuwaiti subject-matter experts to address the OC gap; (3) Moroccan and Kuwaiti school-curriculum item collection to address the IO/IC coverage gaps for those deployment countries; (4) a learner-register Arabic robustness evaluation (none exists [WEB-6, gap\_id 5]) to address the IF gap; and (5) explicit perspective-flagging evaluation specific to the eight-country contested-content scope, which no existing benchmark targets.

| ID  | Evidence                                                                                                    |
|-----|-------------------------------------------------------------------------------------------------------------|
| Q44 | "This 96% score is considered to represent the maximum score meaningfully achievable for ArabicMMLU." (p.4) |

|              |                                                                                                                                                                 |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q71</b>   | "This could be attributed to the relatively small portion (i.e., 6%) of university-level questions in ArabicMMU, which potentially skews the results." (p.7)    |
| <b>WEB-5</b> | 'Beyond MCQ' Arabic open-ended benchmark — supplementary alternative for OF ( <a href="https://arxiv.org/abs/2510.24328">https://arxiv.org/abs/2510.24328</a> ) |
| <b>WEB-6</b> | AraGen 3C3H LLM-as-judge framework for open-ended Arabic generation ( <a href="https://arxiv.org/abs/2510.13430">https://arxiv.org/abs/2510.13430</a> )         |

# Input Ontology (IO)

Score: 3/5 — Moderate gaps | Confidence: high

**Benchmark:** ARABICMLU | **Context:** Non-Arab Tourists and Expats in Eight Arabic-Speaking Countries (ArabicMLU Deployment)

**Input Ontology definition:** Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

The benchmark's taxonomy of 40 tasks across school subjects and education levels [Q25] aligns well with the deployment's school-curriculum knowledge scope (history, geography, Arabic language, civics) [Q72, Q73]. However, several taxonomic issues materially harm content validity for this deployment: (a) the taxonomy includes categories irrelevant to the tourist/expat school-knowledge use case — Driving Test, professional Law (Morocco-only), vocational General Knowledge — that were observed in the sampled data [DATASET-D5, DATASET-D26, DATASET-D27]; (b) university-level content is only 6% of the dataset [Q71]; (c) Civics content as instantiated is overwhelmingly Jordanian-specific rather than pan-Arab [DATASET-D4]; (d) Morocco's school-curriculum (history/geography/civics) is essentially absent from the sample, with Morocco's contribution concentrated in professional law [DATASET-D5]; (e) Kuwait does not appear at all in the sample [Limitations.5]. The future-work acknowledgment of needed extensions — short-answer questions, larger regional coverage, professional-domain content [Q88] — confirms these are recognized gaps.

| ID  | Evidence                                                                                                                                                                                                               |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q25 | "ArabicMLU is an Arabic multiple-choice question-answering dataset comprising 40 tasks spanning a wide range of subjects and education levels." (p.3)                                                                  |
| Q71 | "This could be attributed to the relatively small portion (i.e., 6%) of university-level questions in ArabicMMU, which potentially skews the results." (p.7)                                                           |
| Q72 | "We present the performance of open-source models on selected subjects that potentially contain Arabic-specific contexts." (p.7)                                                                                       |
| Q73 | "These subjects include history, geography, civics, political" (p.7)                                                                                                                                                   |
| Q88 | "For future work, ArabicMLU can be extended to include short-answer or essay questions, different modalities (i.e., images, audio, video), larger region coverage, and more questions in professional settings." (p.9) |
| D4  | لُدُع...يندرأل ايبرعلال شيجلال تالايكشت نم فيبرعلال قوقل... 1952م اندرأل اي في فيفالتئا ةموكلح لوأ ليكشت مت — All 5 sampled Civics Middle School items are exclusively about Jordanian political history               |
| D5  | ين طول اعافدل اب ةفل كل الم ةي موكحل ال طلسل او لدعل ريزو نم... فانئتسالا لجأ ئدتبي (Professional) items are from Morocco (Moroccan procedural law)                                                                    |
| D26 | رتم 50 دعب لىل ع؟عضو كليل نيأ ،سكاعال ريذحتل ثلثم كي دلو كتبكرم ل طعت ل ا ح في — Driving test questions from UAE, Lebanon, Egypt — practical/regulatory, not school-curriculum history/geography                       |
| D27 | هتياهن نم بولسم لكشل اي بولول رامسم :وه جاصلال نس يغرب — Jordan primary GK: sheet metal screw definition — vocational/practical knowledge                                                                              |

## Strengths

- Authentic school-curriculum framing across primary/middle/high school directly matches the deployment's knowledge scope [Q25, Q28, DATASET-D38, DATASET-D34]
- Subject groups explicitly include Arabic-specific contexts (history, geography, civics, political science) that are deployment-relevant [Q72, Q73]
- Broad multi-subject taxonomy with level-appropriate complexity confirmed in sampled data [DATASET-D38, DATASET-D22, DATASET-D34]

| ID | Evidence |
|----|----------|
|----|----------|

|     |                                                                                                                                                                                                                         |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q25 | "ArabicMMLU is an Arabic multiple-choice question-answering dataset comprising 40 tasks spanning a wide range of subjects and education levels." (p.3)                                                                  |
| Q28 | "The subjects are drawn from different education levels (primary school, middle school, and KSA, prioritize Islamic studies alongside subjects like mathematics, natural science, social science, and geography." (p.3) |
| Q72 | "We present the performance of open-source models on selected subjects that potentially contain Arabic-specific contexts." (p.7)                                                                                        |
| Q73 | "These subjects include history, geography, civics, political" (p.7)                                                                                                                                                    |
| D22 | قيعبتل اب ةمئل س نوك ت لي غشت ل ةلمع نإف ... ةمئل س تال خدم ل ت ناك اذا هنأ بسا ح ل لوح ةعجارمل بول س أ ضررت في<br>— Egypt university accounting — auditing/financial statements questions                              |
| D34 | ةرونم ل ةني دمل : ل مئ دقل م سال وه برئي — Jordan primary history: ancient name of Medina — Islamic history, factual                                                                                                    |
| D38 | + موي سني غام ل ؟ ل ص ح ي ذل ل لعافت ل فص ت ةي ل ات ل تال داعم ل أ ؛ رب ت خ م ل اب موي سني غام ل ن م ةي م ك قرح ب دم ح أ م اق<br>— Natural science MS: chemistry reactions — country-neutral STEM                       |

## Checklist

### Checklist item definitions:

| ID   | Assessment Question                                            |
|------|----------------------------------------------------------------|
| IO-1 | Identify test case categories required for regional deployment |
| IO-2 | Check if taxonomy omits regionally relevant categories         |
| IO-3 | Check if taxonomy includes irrelevant categories               |
| IO-4 | Document category gaps harming content validity                |

### Checklist responses:

**IO-1:** Required categories for this deployment: Arabic language, Arab history, Arab geography, civics/social science at school-curriculum level [elicitation Q1, A1]; explicitly out of scope are dialectal phrases, tourist etiquette, practical travel, and professional domains.

**IO-2:** Yes — country-specific school-curriculum coverage is unequal. Morocco's school subjects are largely absent from the sample (only professional law observed) [DATASET-D5]; Kuwait does not appear at all [Limitations.5]; civics is Jordan-only in the sample [DATASET-D4]. The paper acknowledges unequal country representation [Q89, Q90, Q91].

**IO-3:** Yes — Driving Test [DATASET-D26], professional Law [DATASET-D5], vocational General Knowledge (sheet metal screws, hospital names) [DATASET-D27, DATASET-D10], and University Accounting/Management [DATASET-D22, DATASET-D23] are out of scope for the deployment's school-curriculum tourist knowledge use case.

**IO-4:** Documented gaps: (1) absence of Maghrebi school-curriculum content for Morocco; (2) absence of Kuwaiti curriculum; (3) lack of a multi-perspective or contested-content taxonomic dimension despite the subjects most relevant to the deployment being inherently contested [Q72, Q73]; (4) university-level content underrepresented at 6% [Q71].

| ID  | Evidence                                                                                                                                                                                                                                                  |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q1  | "we present ArabicMMLU, the first multi-task language understanding benchmark for the Arabic language, sourced from school exams across diverse educational levels in different countries spanning North Africa, the Levant, and the Gulf regions." (p.1) |
| Q25 | "ArabicMMLU is an Arabic multiple-choice question-answering dataset comprising 40 tasks spanning a wide range of subjects and education levels." (p.3)                                                                                                    |
| Q71 | "This could be attributed to the relatively small portion (i.e., 6%) of university-level questions in ArabicMMU, which potentially skews the results." (p.7)                                                                                              |
| Q72 | "We present the performance of open-source models on selected subjects that potentially contain Arabic-specific contexts." (p.7)                                                                                                                          |
| Q73 | "These subjects include history, geography, civics, political" (p.7)                                                                                                                                                                                      |
| Q88 | "For future work, ArabicMMLU can be extended to include short-answer or essay questions, different modalities (i.e., images, audio, video), larger region coverage, and more questions in professional settings." (p.9)                                   |
| Q89 | "ArabicMMLU does not represent all Arabic countries equally. For example, we have collected over 6K multiple-choice questions from Jordan, while other countries are represented with only 100 questions or, in some cases, not at all." (p.9)            |

|       |                                                                                                                                                                                                        |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q90   | "This is largely due to the availability of publicly-accessible exams in each country; some countries have digitized their exams, but not others." (p.9)                                               |
| Q91   | "Additionally, our search for relevant Arabic content across the internet was not exhaustive." (p.9)                                                                                                   |
| D4    | ل دُع...يندرأال يبرعلال شيجال تاليكشت نم دبيبرعلال ةوقلا... 1952م اعدرأال يف ةيفالائا ةموكح لوأ ليكشت مت<br>All 5 sampled Civics Middle School items are exclusively about Jordanian political history |
| D5    | ينطولاعافدلابل ةفلكملالميموكحال ةطلسلالو لدعلال ريزو نم...فانئسالا لجأ ئدتبي<br>(Professional) items are from Morocco (Moroccan procedural law)                                                        |
| D10   | Jordan — ةيبطلال ةني دملال يفتشتسم...ةصصختمل تاسسؤمل نم ديدعلالميموكحال ةيكلملالميبطلال تامدخال لمشت<br>Jordan primary social science: Jordan Royal Medical Services institutions                      |
| D22   | ةيعبتلاب ةميسللس نوكت ليغشتال ةيلمع نإف...ةميسللس تالخدمال تناك اذا هنأ بساحال لوح ةعجارملال بولسأ ضررتفي<br>Egypt university accounting — auditing/financial statements questions                     |
| D23   | Severely garbled OCR text making question almost illegible — ة زكرم لا: لي ريتذت...صاختلأال نيتب ةطلتدل عتييهت ةتجرد                                                                                   |
| D26   | Driving test questions from UAE, Lebanon, Egypt — رتم 50 دعب لىل ع؟هعضو كليلع نيأ ،سكاعال ريذحتال ثلثم كي دلو كتبكرمل طعت لاج يف<br>practical/regulatory, not school-curriculum history/geography      |
| D27   | Jordan primary GK: sheet metal screw definition — هتياهن نم بولسم لكشلال يبلول رامسم: وه جاصلال نس يغرب<br>vocational/practical knowledge                                                              |
| WEB-3 | Palestine confirmed top-3 source country — yet no Palestinian annotators in pool ( <a href="https://github.com/mbzuai-nlp/ArabicMMLU">https://github.com/mbzuai-nlp/ArabicMMLU</a> )                   |

### Information Gaps

- Exact per-country, per-subject distribution counts not extractable from documentation
- Whether Country=null items are truly pan-Arab or reflect missing metadata

### Requires Expert Verification

- Whether sub-regional curriculum distinctions (Maghrebi, Levantine, Gulf) are meaningfully represented in the full dataset beyond the sampled configs

### Remediation

**Gap 1: Out-of-scope categories (Driving Test, professional Law, vocational General Knowledge) inflate aggregate scores irrelevantly [DATASET-D5, DATASET-D26, DATASET-D27]**

**Recommendation:** Define a deployment-relevant subset of ArabicMMLU configs (history, geography, civics, Arabic language, social science, Islamic studies at appropriate framing) and report scores only on this subset; exclude driving test, professional law, and vocational items.

| ID  | Evidence                                                                                                                                                                                          |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D5  | All 5 sampled Law (Professional) items are from Morocco (Moroccan procedural law) — ينطولاعافدلابل ةفلكملالميموكحال ةطلسلالو لدعلال ريزو نم...فانئسالا لجأ ئدتبي                                  |
| D26 | Driving test questions from UAE, Lebanon, Egypt — رتم 50 دعب لىل ع؟هعضو كليلع نيأ ،سكاعال ريذحتال ثلثم كي دلو كتبكرمل طعت لاج يف<br>practical/regulatory, not school-curriculum history/geography |
| D27 | Jordan primary GK: sheet metal screw definition — هتياهن نم بولسم لكشلال يبلول رامسم: وه جاصلال نس يغرب<br>vocational/practical knowledge                                                         |

# Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

**Benchmark:** ARABICMMLU | **Context:** Non-Arab Tourists and Expats in Eight Arabic-Speaking Countries (ArabicMMLU Deployment)

**Input Content definition:** Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

Authentic ground-up sourcing from school exams across eight countries [Q1, Q26] is a substantial strength over translation-based prior work [Q5, Q6], and Palestine being a top-3 source country is materially better than the scaffold anticipated [WEB-3, WEB-4]. However, content distribution is severely unequal: Jordan contributes 6,000+ questions while some countries contribute as few as 100 [Q89]; Morocco lacks annotators and Maghrebi school-curriculum coverage [DATASET-D5]; Kuwait is effectively absent [Limitations.5]; and contested historical content is framed through Jordanian curriculum perspective [DATASET-D1, DATASET-D2, DATASET-D3]. OCR corruption in some Egyptian university items [DATASET-D23, DATASET-D24] introduces construct-irrelevant variance. Islamic Studies content is presented prescriptively from an internal Muslim perspective [DATASET-D14, DATASET-D35], misaligned with the non-Muslim tourist target population. Per the elicitation, pan-Arab framing is acceptable [A2], but the system should flag country-specific content — yet the benchmark's ground-truth treats Jordanian framing as the silent default for civic and historical questions [DATASET-D4].

| ID  | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q1  | "we present ArabicMMLU, the first multi-task language understanding benchmark for the Arabic language, sourced from school exams across diverse educational levels in different countries spanning North Africa, the Levant, and the Gulf regions." (p.1)                                                                                                                                                                                                                                                                                                                                     |
| Q5  | "Although large language models (LLMs) such as GPT-3.5 (Ouyang et al., 2022), BLOOMZ (Muennighoff et al., 2022), and Jais (Sengupta et al., 2023) have been pretrained with substantial coverage of Modern Standard Arabic (MSA), their reasoning and knowledge assessments are primarily conducted using datasets translated from English to Arabic (Sengupta et al., 2023; Huang et al., 2023), which means there is limited capacity to evaluate content specific to Arabic." (p.1)                                                                                                        |
| Q6  | "This reliance on translation systems not only demonstrates an Anglocentric approach (Ramesh et al., 2023; Talat et al., 2022) but also potentially introduces errors and biases." (p.1)                                                                                                                                                                                                                                                                                                                                                                                                      |
| Q26 | "The questions are sourced from eight different countries in North Africa (Morocco and Egypt), the Levant (Jordan, Palestine, and Lebanon), and the Gulf (UAE, Kuwait, and KSA)." (p.3)                                                                                                                                                                                                                                                                                                                                                                                                       |
| Q89 | "ArabicMMLU does not represent all Arabic countries equally. For example, we have collected over 6K multiple-choice questions from Jordan, while other countries are represented with only 100 questions or, in some cases, not at all." (p.9)                                                                                                                                                                                                                                                                                                                                                |
| D1  | دحو مامت؛ ني طسلف يف اهعاطأو ةينوي هصلا ةكرحلا نم لوألا هلل ادبع كل مل فق اوام نم تسيل ةيتآلا نم دحاو هضف / ني طسلف ةمألا قودنص معد يل برعلا ةودع / قاربلا طئاح يل ارمم دوهيلا اعاطع ا هضف / م 1950 اع ني تفضلا اعاطع ا هضف / ني طسلف ةمألا قودنص معد يل برعلا ةودع / قاربلا طئاح يل ارمم دوهيلا اعاطع ا هضف / م 1950 اع ني تفضلا اعاطع ا هضف / ني طسلف ةمألا قودنص معد يل برعلا ةودع / قاربلا طئاح يل ارمم دوهيلا اعاطع ا هضف / م 1950 اع ني تفضلا<br>— Jordan HS history question about King Abdullah I's positions on Zionism and Palestine — framed from Jordanian curriculum perspective |
| D2  | نب لصل يف ري مأل ٢ / ني سحلا نب هلل ادبع ري مأل ١: سي راب يف ءافلحل هديق يذل ا حلصل رم تؤم يف برعلا _____ لثم<br>— Jordan civics HS: who represented Arabs at Paris Peace Conference — Jordanian Hashemite framing                                                                                                                                                                                                                                                                                                                                                                            |
| D3  | Jordan civics HS: بادتالال كف دويقي نم دالبل ررحت: ١٩٢٨ ماع هي ناطريبل هي ندرألا هدهاع ملل سيئرل فدهلا ناك<br>— primary goal of the 1948 Jordanian-British treaty — Jordanian national framing                                                                                                                                                                                                                                                                                                                                                                                                |
| D4  | لدُع... يندرألا ي برعلا شيجل تالليكشت نم ةبرعلا ةوق... 1952 ماع ندرألا يف ةيفالائا ةموكح لوألي لكشت مت<br>— All 5 sampled Civics Middle School items are exclusively about Jordanian political history                                                                                                                                                                                                                                                                                                                                                                                        |
| D5  | All 5 sampled Law (Professional) items are from Morocco (Moroccan procedural law)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| D14 | — اتارمع عربأ؟ لمسو هيلع هلل يلص ي ب نل رم نعا ةرم مك / ماسلا هيلع نامي لس؟ ريطلا قطنم ملع يذل ي ب نل وه نم<br>— Pan-Islamic knowledge questions: no country field — factual hadith/Quran questions                                                                                                                                                                                                                                                                                                                                                                                           |

|       |                                                                                                                                                                                      |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D23   | Severely garbled OCR text making question almost illegible                                                                                                                           |
| D24   | Egypt university political science with severe OCR corruption — content partially unreadable                                                                                         |
| D35   | Arabic language reading with Islamic ritual context — assumes Muslim reader perspective                                                                                              |
| WEB-3 | Palestine confirmed top-3 source country — yet no Palestinian annotators in pool ( <a href="https://github.com/mbzuai-nlp/ArabicMMLU">https://github.com/mbzuai-nlp/ArabicMMLU</a> ) |
| WEB-4 | HuggingFace dataset card confirms Palestine major contributor ( <a href="https://huggingface.co/datasets/MBZUAI/ArabicMMLU">https://huggingface.co/datasets/MBZUAI/ArabicMMLU</a> )  |

## Strengths

- Ground-up data collection from authentic school exams in eight countries, distinct from prior translation-based approaches [Q1, Q5, Q6, Q26]
- Palestine confirmed as top-3 source country with substantive curriculum content across multiple subjects [WEB-3, WEB-4, DATASET-D17, DATASET-D20, DATASET-D21]
- Native-speaker workforce with regional collaborators from Jordan, Egypt, Lebanon, UAE, and Saudi Arabia [Q31, Q96]
- GPT-4 performance gap (80% on translated MMLU vs 72.5% on ArabicMMLU) supports construct validity for culturally grounded knowledge [Q64, Q65]

| ID    | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q1    | “we present ArabicMMLU, the first multi-task language understanding benchmark for the Arabic language, sourced from school exams across diverse educational levels in different countries spanning North Africa, the Levant, and the Gulf regions.” (p.1)                                                                                                                                                                                                                              |
| Q5    | “Although large language models (LLMs) such as GPT-3.5 (Ouyang et al., 2022), BLOOMZ (Muennighoff et al., 2022), and Jais (Sengupta et al., 2023) have been pretrained with substantial coverage of Modern Standard Arabic (MSA), their reasoning and knowledge assessments are primarily conducted using datasets translated from English to Arabic (Sengupta et al., 2023; Huang et al., 2023), which means there is limited capacity to evaluate content specific to Arabic.” (p.1) |
| Q6    | “This reliance on translation systems not only demonstrates an Anglocentric approach (Ramesh et al., 2023; Talat et al., 2022) but also potentially introduces errors and biases.” (p.1)                                                                                                                                                                                                                                                                                               |
| Q26   | “The questions are sourced from eight different countries in North Africa (Morocco and Egypt), the Levant (Jordan, Palestine, and Lebanon), and the Gulf (UAE, Kuwait, and KSA).” (p.3)                                                                                                                                                                                                                                                                                                |
| Q31   | “The data construction process involved a total of 10 Arabic native speakers from different countries: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 from UAE, and 2 from KSA) and 4 external workers (3 Jordanian and 1 Egyptian).” (p.4)                                                                                                                                                                                                                                |
| Q64   | “It is noteworthy that in the GPT-4 technical report (OpenAI, 2023), the accuracy of the English-Arabic translated MMLU dataset is reported as 80%, which is 8 points higher than our ArabicMMU results.” (p.7)                                                                                                                                                                                                                                                                        |
| Q65   | “One possible explanation for this difference is that our ArabicMMU presents a greater challenge due to its inclusion of a higher proportion of Arabic-specific content.” (p.7)                                                                                                                                                                                                                                                                                                        |
| Q96   | “We extend our gratitude to all collaborators from Jordan, Egypt, Lebanon, UAE, and Saudi Arabia who participated in the data collection process.” (p.10)                                                                                                                                                                                                                                                                                                                              |
| D17   | Palestine middle school geography: world geography (Oceania islands) — no regional specificity                                                                                                                                                                                                                                                                                                                                                                                         |
| D20   | Palestine primary CS: binary number systems — STEM, no contested content                                                                                                                                                                                                                                                                                                                                                                                                               |
| D21   | Palestine HS biology: standard cellular biology — culturally neutral STEM                                                                                                                                                                                                                                                                                                                                                                                                              |
| WEB-3 | Palestine confirmed top-3 source country — yet no Palestinian annotators in pool ( <a href="https://github.com/mbzuai-nlp/ArabicMMLU">https://github.com/mbzuai-nlp/ArabicMMLU</a> )                                                                                                                                                                                                                                                                                                   |
| WEB-4 | HuggingFace dataset card confirms Palestine major contributor ( <a href="https://huggingface.co/datasets/MBZUAI/ArabicMMLU">https://huggingface.co/datasets/MBZUAI/ArabicMMLU</a> )                                                                                                                                                                                                                                                                                                    |

## Checklist

### Checklist item definitions:

| ID | Assessment Question |
|----|---------------------|
|----|---------------------|

|      |                                                                            |
|------|----------------------------------------------------------------------------|
| IC-1 | Do inputs require region-specific cultural/geographic/dialectal knowledge? |
| IC-2 | Does culturally sensitive content align with target culture?               |
| IC-3 | Flag inputs assuming Western-specific knowledge                            |
| IC-4 | Regional annotator recruitment for culturally sensitive instances          |
| IC-5 | Document content issues harming content validity                           |

### Checklist responses:

**IC-1:** Yes — many inputs require region-specific knowledge of Arab history, geography, Jordanian/Egyptian/Palestinian curricula, and Islamic religious practice [Q72, Q73, DATASET-D1, DATASET-D14].

**IC-2:** Partially — pan-Arab framing is acceptable per elicitation [A2], but contested historical content (e.g., King Abdullah I and Zionism [DATASET-D1], 1948 events) is framed from Jordanian curriculum perspective without flagging this country-specificity, conflicting with the deployment requirement to flag rather than silently generalize country-specific content.

**IC-3:** Limited Western-specific content concern — some Geography items ask about non-Arab global facts (Kilimanjaro, Shanghai) [MINOR Concern 11], but these are general world knowledge rather than Western-specific cultural content. The more pressing issue is Jordan-specific content presented as default Arab content.

**IC-4:** Verified by 2 native Arabic speakers on 100 sampled questions [Q42], yielding 96% match [Q43] — but no Moroccan, Palestinian, or Kuwaiti annotators participated in collection or verification [Q31, Q96]. Palestinian-sourced contested content was verified by non-Palestinian workers.

**IC-5:** Documented content issues: (1) Jordan-skew (~6,000+ questions) vs. ~100 for others [Q89]; (2) Morocco's school-curriculum content largely absent [DATASET-D5]; (3) Kuwait effectively absent [Limitations.5]; (4) OCR corruption in university Egyptian items [DATASET-D23, DATASET-D24]; (5) Islamic Studies presented prescriptively for Muslim students rather than explanatorily for non-Muslim visitors [DATASET-D14, DATASET-D35]; (6) Jordanian-curriculum framing of contested historical events [DATASET-D1, DATASET-D2, DATASET-D3].

| ID  | Evidence                                                                                                                                                                                                                                                                                                                                                                                   |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q31 | "The data construction process involved a total of 10 Arabic native speakers from different countries: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 from UAE, and 2 from KSA) and 4 external workers (3 Jordanian and 1 Egyptian)." (p.4)                                                                                                                                    |
| Q42 | "Finally, we assessed the accuracy of our data collection by having two native Arabic speakers annotate 100 randomly sampled questions. They were provided with all metadata, including the answer key, and tasked with verifying the correctness of each sample using any available resources (e.g., search engines)." (p.4)                                                              |
| Q43 | "We found that 96% of the questions and answer keys match on average, while the remaining could have incorrect answer keys." (p.4)                                                                                                                                                                                                                                                         |
| Q72 | "We present the performance of open-source models on selected subjects that potentially contain Arabic-specific contexts." (p.7)                                                                                                                                                                                                                                                           |
| Q73 | "These subjects include history, geography, civics, political" (p.7)                                                                                                                                                                                                                                                                                                                       |
| Q89 | "ArabicMMLU does not represent all Arabic countries equally. For example, we have collected over 6K multiple-choice questions from Jordan, while other countries are represented with only 100 questions or, in some cases, not at all." (p.9)                                                                                                                                             |
| Q96 | "We extend our gratitude to all collaborators from Jordan, Egypt, Lebanon, UAE, and Saudi Arabia who participated in the data collection process." (p.10)                                                                                                                                                                                                                                  |
| D1  | دحو مامت! :نيطسلف يف اعامطأو ةينويهصل ةكرحلا نم لوأل هللادبع لكلمل فقاوم نم تسيل ةيتأل نم ةدحاو هضفر / نيطسلف ةمأل قودنص معد إلى برعلا ةودع / قاربلا طئاح إلى ارمم دوهيلا اعطاع هضفر / م 1950 اع ني تفضلا اعطاق اضا فر روفل دعوو وكيب سكي اس ةدهاعم — Jordan HS history question about King Abdullah I's positions on Zionism and Palestine — framed from Jordanian curriculum perspective |
| D2  | نب لصيف ريملأل ٢ / نيسحلا نب هللادبع ريملأل ١: سيراب يف ءافلحل اهدق يف اذل حلصل رمتؤم يف برعلا. لثم — Jordan civics HS: who represented Arabs at Paris Peace Conference — Jordanian Hashemite framing                                                                                                                                                                                      |
| D3  | Jordan civics HS: بادتالال كف دويقي نم دالبل ررحت: ١٩٢٨ ماع هي ناطريبل هي ندرألا هدهاعم لل سيئرل فدهل ناك — primary goal of the 1948 Jordanian-British treaty — Jordanian national framing                                                                                                                                                                                                 |
| D5  | ين طول اعافدل اب ءفل كملم ءي موكحلا طلسل او لدعل ريزو نم ... افانئستالال لجا ئدتبي (Professional) items are from Morocco (Moroccan procedural law)                                                                                                                                                                                                                                         |

|     |                                                                                                                                                                                                      |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D14 | — تارمع عرباً؟ لم يسو هيلع هلل لى لص يبنل ارم تع ارم كم / مالسلال هيلع نامي لس ؟ ري طلا قطنم ملع يذل يبنل وه نم — Pan-Islamic knowledge questions: no country field — factual hadith/Quran questions |
| D23 | — Severely garbled OCR text making question almost illegible                                                                                                                                         |
| D24 | — Egypt university political science with severe OCR corruption — content partially unreadable                                                                                                       |
| D35 | — Arabic language reading with Islamic ritual context — assumes Muslim reader perspective                                                                                                            |

### Information Gaps

- Exact per-country question counts beyond top-3 not published
- Proportion of OCR-corrupted items in full dataset

### Requires Expert Verification

- Whether ground-truth labels for Palestinian-sourced contested-history questions reflect Palestinian historiography
- Whether Moroccan curriculum perspectives in history/geography are present in the full dataset beyond the sampled configs
- Whether Islamic Studies items would be considered appropriately framed for non-Muslim educational use

### Remediation

**Gap 1: Morocco's school-curriculum content (history/geography/civics) is largely absent; Kuwait is effectively absent [DATASET-D5, Limitations.5]**

**Recommendation:** Commission targeted collection of Moroccan and Kuwaiti primary/middle/high school exam items in history, geography, and civics, with native annotators from each country, to extend country coverage before deployment-relevant evaluation.

| ID | Evidence                                                                            |
|----|-------------------------------------------------------------------------------------|
| D5 | — All 5 sampled Law (Professional) items are from Morocco (Moroccan procedural law) |

# Input Form (IF)

Score: 3/5 — Moderate gaps | Confidence: high

**Benchmark:** ARABICMMLU | **Context:** Non-Arab Tourists and Expats in Eight Arabic-Speaking Countries (ArabicMMLU Deployment)

**Input Form definition:** Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

The benchmark is uniformly text-only MSA [Q2, Q92], aligning with the deployment's text-only modality [A4] and MSA target [elicitation]. Multimodal content was discarded [Q37] and contextual passages preserved where needed [Q38]. However, the benchmark's MSA is fluent native-speaker register from school exams, while the deployment's users are non-native learners with basic-to-intermediate MSA proficiency, likely producing grammatically imperfect or code-switched Arabic [elicitation, expected\_arabic\_proficiency]. This signal-distribution gap is structural and undocumented as a design trade-off in the paper. The benchmark also notes that real-world Arabic LLM exposure includes dialectal Arabic, which is excluded here [Q92]. One minor data-quality issue observed: an Arabic Language Grammar item with English question stem [DATASET-D16] inconsistent with the stated Arabic-only design [Q30].

| ID  | Evidence                                                                                                                                                                                                            |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q2  | "Our data comprises 40 tasks and 14,575 multiple-choice questions in Modern Standard Arabic (MSA) and is carefully constructed by collaborating with native speakers in the region." (p.1)                          |
| Q30 | "When designing ArabicMMLU, we excluded questions in English and only included questions in Arabic." (p.3)                                                                                                          |
| Q37 | "During manual data scraping, workers were instructed to include only questions accompanied by an answer key, and to discard questions containing multi-modal information (e.g., images, videos, or tables)." (p.4) |
| Q38 | "If a question had additional contextual information (e.g., a passage referenced by several questions), the context was required to be included with each corresponding question." (p.4)                            |
| Q92 | "The dataset primarily focuses on Modern Standard Arabic (MSA). However, multilingual and Arabic LLMs are often exposed to both MSA and dialectal Arabic." (p.9)                                                    |
| D16 | In the following Quranic verse, what is the correct parsing of the word كَلْبٌ — Grammar question with English question stem — unexpected language mixing in nominally Arabic benchmark                             |

## Strengths

- Text-only MSA format directly aligns with deployment's text-based modality [Q2, Q93, A4]
- Multimodal content systematically excluded; contextual passages preserved for dependent questions [Q37, Q38]
- Prompt language variants (Arabic/English) tested, providing some signal on cross-lingual prompt sensitivity [Q47]
- Uniform MSA register confirmed across all sampled configs [DATASET-D29, DATASET-D15]

| ID  | Evidence                                                                                                                                                                                                                                                                                   |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q2  | "Our data comprises 40 tasks and 14,575 multiple-choice questions in Modern Standard Arabic (MSA) and is carefully constructed by collaborating with native speakers in the region." (p.1)                                                                                                 |
| Q37 | "During manual data scraping, workers were instructed to include only questions accompanied by an answer key, and to discard questions containing multi-modal information (e.g., images, videos, or tables)." (p.4)                                                                        |
| Q38 | "If a question had additional contextual information (e.g., a passage referenced by several questions), the context was required to be included with each corresponding question." (p.4)                                                                                                   |
| Q47 | "We initially conducted experiments with four settings: (1) Arabic prompt and Arabic alphabetic output, (2) Arabic prompt and English (i.e. Latin script) alphabetic output, (3) English prompt and Arabic alphabetic output, and (4) English prompt and English alphabetic output." (p.5) |

|     |                                                                                                                                                                                                                   |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q93 | “ArabicMMLU is focused solely on text-based assessment, and the exploration of multimodal questions is left for future work.” (p.9)                                                                               |
| D15 | لإمّش لآ. [أارف] برعب نوي نان دعل آي مّس... نينس لآ لى ل برعل خيرات دوعي — Pan-Arab ethnolinguistic history reading comprehension — culturally grounded, no country marker                                        |
| D29 | بصنو دي كوت فرح نّ قمل كل حي حص لآ بارعل لآ وه ام، [أتاح لآصل لآ أول مّعو أو نّ مّآ نّ لآ نّ] قني نّ أرق لآ قني لآ ي ف — Arabic grammar parsing using Quranic verse — MSA grammar directly relevant to deployment |

## Checklist

### Checklist item definitions:

| ID   | Assessment Question                                          |
|------|--------------------------------------------------------------|
| IF-1 | Compare signal distributions between source and target       |
| IF-2 | Check if regional infrastructure supports data capture specs |
| IF-3 | Identify domain-specific form differences                    |
| IF-4 | Document form mismatches harming external validity           |

### Checklist responses:

**IF-1:** Signal distribution mismatch: benchmark assumes fluent native-speaker MSA from school exams [Q2], while deployment users produce learner-register MSA potentially with grammatical imperfection and code-switching [elicitation expected\_arabic\_proficiency, languages.user\_input\_register]. No dedicated benchmark for non-native Arabic robustness identified [WEB-6, gap\_id 5].

**IF-2:** Infrastructure supports text-only Arabic input; RTL rendering requirement noted [writing\_systems]. No infrastructure barriers identified for tourists/expats with personal devices [infrastructure\_notes].

**IF-3:** Domain-specific form differences: dialectal Arabic excluded from benchmark [Q92] but is common in real-world MENA usage; learner code-switching with Latin script likely [writing\_systems] but not represented.

**IF-4:** Form mismatch: learner-register input vs. fluent-MSA evaluation creates undocumented signal-distribution gap. One data-quality artifact: English question stem in nominally Arabic-only benchmark [DATASET-D16].

| ID    | Evidence                                                                                                                                                                                   |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q2    | “Our data comprises 40 tasks and 14,575 multiple-choice questions in Modern Standard Arabic (MSA) and is carefully constructed by collaborating with native speakers in the region.” (p.1) |
| Q30   | “When designing ArabicMMLU, we excluded questions in English and only included questions in Arabic.” (p.3)                                                                                 |
| Q92   | “The dataset primarily focuses on Modern Standard Arabic (MSA). However, multilingual and Arabic LLMs are often exposed to both MSA and dialectal Arabic.” (p.9)                           |
| Q93   | “ArabicMMLU is focused solely on text-based assessment, and the exploration of multimodal questions is left for future work.” (p.9)                                                        |
| D16   | In the following Quranic verse, what is the correct parsing of the word كَب — Grammar question with English question stem — unexpected language mixing in nominally Arabic benchmark       |
| WEB-6 | AraGen 3C3H LLM-as-judge framework for open-ended Arabic generation (https://arxiv.org/abs/2510.13430)                                                                                     |

## Information Gaps

- No empirical evaluation of model performance on learner-register or code-switched Arabic input

## Requires Expert Verification

- Whether tourist/expat learner Arabic deviates from benchmark MSA in ways that materially affect downstream system behavior

## Remediation

**Gap 1: Benchmark MSA register does not represent learner-register, code-switched Arabic produced by non-native users [Q92, elicitation expected\_arabic\_proficiency]**

**Recommendation:** Construct a small bespoke evaluation set of learner-register and code-switched Arabic queries paraphrasing ArabicMMLU items, and measure performance degradation; this addresses the documentation gap noted in [WEB-6] where no learner-register Arabic benchmark exists.

| ID    | Evidence                                                                                                                                                         |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q92   | "The dataset primarily focuses on Modern Standard Arabic (MSA). However, multilingual and Arabic LLMs are often exposed to both MSA and dialectal Arabic." (p.9) |
| WEB-6 | AraGen 3C3H LLM-as-judge framework for open-ended Arabic generation ( <a href="https://arxiv.org/abs/2510.13430">https://arxiv.org/abs/2510.13430</a> )          |

# Output Ontology (OO)

Score: 1/5 — Serious concern | Confidence: high

**Benchmark:** ARABICMMLU | **Context:** Non-Arab Tourists and Expats in Eight Arabic-Speaking Countries (ArabicMMLU Deployment)

**Output Ontology definition:** Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

This is the most critical structural mismatch for this deployment. The label schema is exclusively single-correct-answer MCQ with 2–5 candidate options [Q27], and every sampled example confirms this format [DATASET-D1, all]. The deployment explicitly requires multi-perspective acknowledgment of contested historical and civic questions [elicitation A3], but the benchmark's output taxonomy structurally cannot represent or score such pluralistic responses. The four prompt-configuration variants tested vary only in prompt language and output script [Q47], not in label structure. Contested questions like King Abdullah I and Zionism [DATASET-D1], GCC founding states [DATASET-D25], and 'elements of a state' with 'occupation' as a distractor option [DATASET-D18] are all reduced to single correct labels reflecting one curriculum's framing. The future-work note proposing short-answer and essay questions [Q88] tacitly acknowledges this limitation. Open-ended Arabic evaluation alternatives exist (AraGen, 'Beyond MCQ' — [WEB-5, WEB-6]) but are not part of ArabicMMLU itself.

| ID    | Evidence                                                                                                                                                                                                                                                                                                                                                                                              |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q27   | "Each question has 2–5 candidate answers, with one correct answer." (p.3)                                                                                                                                                                                                                                                                                                                             |
| Q47   | "We initially conducted experiments with four settings: (1) Arabic prompt and Arabic alphabetic output, (2) Arabic prompt and English (i.e. Latin script) alphabetic output, (3) English prompt and Arabic alphabetic output, and (4) English prompt and English alphabetic output." (p.5)                                                                                                            |
| Q88   | "For future work, ArabicMMLU can be extended to include short-answer or essay questions, different modalities (i.e., images, audio, video), larger region coverage, and more questions in professional settings." (p.9)                                                                                                                                                                               |
| D1    | دحو مامتإ: ني طسلف يف اهعماطأو ةي نوي هصلا ةك رحلا نم لوألا هلل ادبع كل مل فق اوام نم تسيل ةيتآل نم ةدحاو هضفر / ني طسلف ةمأل قودنص معد إلب رعلا ةودع / قاربلا طئاح إلب ارهم دوهيلا اعطاع هضفر / م 1950 ماغ ني تفضللا اعطاق اضفر روفلب دعوو وكيب سكي اس ةدهاعم — Jordan HS history question about King Abdullah I's positions on Zionism and Palestine — framed from Jordanian curriculum perspective |
| D18   | — Palestine MS geography: elements of a state — ةداي سلا / لال تحالال / مي لقالا / بعشلا: ادع ةلودل رصانع نم يلي ام عجم "occupation" appears as a distractors option                                                                                                                                                                                                                                  |
| D25   | — Jordan civics HS: false claim about GCC founding states — correct answer is "No" since Yemen and Iraq were not founders هدمتملا هي برغل تارامال) هي جيجلخل نواعتلا سلجم ةماقلا لي تعس يتي لتا لودلا                                                                                                                                                                                                 |
| WEB-5 | 'Beyond MCQ' Arabic open-ended benchmark — supplementary alternative for OF ( <a href="https://arxiv.org/abs/2510.24328">https://arxiv.org/abs/2510.24328</a> )                                                                                                                                                                                                                                       |
| WEB-6 | AraGen 3C3H LLM-as-judge framework for open-ended Arabic generation ( <a href="https://arxiv.org/abs/2510.13430">https://arxiv.org/abs/2510.13430</a> )                                                                                                                                                                                                                                               |

## Strengths

- MCQ format produces well-defined, reproducible scoring for factual recall tasks [Q27]
- Multiple prompt-configuration variants enable analysis of prompt-language sensitivity [Q47]

| ID  | Evidence                                                                                                                                                                                                                                                                                   |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q27 | "Each question has 2–5 candidate answers, with one correct answer." (p.3)                                                                                                                                                                                                                  |
| Q47 | "We initially conducted experiments with four settings: (1) Arabic prompt and Arabic alphabetic output, (2) Arabic prompt and English (i.e. Latin script) alphabetic output, (3) English prompt and Arabic alphabetic output, and (4) English prompt and English alphabetic output." (p.5) |

## Checklist

### Checklist item definitions:

| ID   | Assessment Question                                          |
|------|--------------------------------------------------------------|
| OO-1 | Review output label categories for regional relevance        |
| OO-2 | Identify missing regionally specific categories              |
| OO-3 | Flag categories encoding non-regional values                 |
| OO-4 | Consider stakeholder-driven taxonomy redesign                |
| OO-5 | Document taxonomy issues harming structural/content validity |

### Checklist responses:

**OO-1:** Output label categories are uniform single-correct MCQ [Q27], reviewed against deployment requirement for multi-perspective acknowledgment [elicitation A3] — fundamental misalignment.

**OO-2:** Missing categories: multi-perspective/multi-correct labels, perspective-flagging labels, explanatory/short-answer outputs, abstention labels for contested content. None present in benchmark [Q27, Q88].

**OO-3:** The single-correct-answer assumption encodes a 'one curriculum framing is correct' value that conflicts with the deployment's pluralistic requirement [elicitation A3]; instances like the King Abdullah I question [DATASET-D1] and 'occupation' distractor [DATASET-D18] make this concrete.

**OO-4:** Stakeholder-driven taxonomy redesign needed for deployment use; supplementary frameworks (AraGen 3C3H, 'Beyond MCQ') identified but not specifically targeted at contested historical content [WEB-5, WEB-6, net new findings].

**OO-5:** Documented: structural validity violation (construct of multi-perspective knowledge cannot be represented), content validity violation (multi-perspective categories absent), and external validity risk (benchmark performance will not predict pluralistic response quality).

| ID    | Evidence                                                                                                                                                                                                                                                                                                                                                     |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q27   | "Each question has 2–5 candidate answers, with one correct answer." (p.3)                                                                                                                                                                                                                                                                                    |
| Q88   | "For future work, ArabicMMLU can be extended to include short-answer or essay questions, different modalities (i.e., images, audio, video), larger region coverage, and more questions in professional settings." (p.9)                                                                                                                                      |
| D1    | دحو ومامت! :نيطسلف يف اهعامطأو قيوئويھصل اءكءرءل نم لوألا هلل ادبع كللم الم بقاوم نم تسيل قئآل نم دءاوا هضفر / ني طسلفل اءمألا قوئنص مءء قلا برعل اءوء / قاربلا طئاح قلا ارمم دوھيلا اعطاء هضفر / م 1950 ماع نيئفضلا<br>Jordan HS history question about King Abdullah I's positions on Zionism and Palestine — framed from Jordanian curriculum perspective |
| D18   | Palestine MS geography: elements of a state — قءاي سلا / اللالءال / ميلاقال / بعشلا : ادع ءولءلا رصانع نم يلى ام عءء — "occupation" appears as a distractors option                                                                                                                                                                                          |
| WEB-5 | 'Beyond MCQ' Arabic open-ended benchmark — supplementary alternative for OF ( <a href="https://arxiv.org/abs/2510.24328">https://arxiv.org/abs/2510.24328</a> )                                                                                                                                                                                              |
| WEB-6 | AraGen 3C3H LLM-as-judge framework for open-ended Arabic generation ( <a href="https://arxiv.org/abs/2510.13430">https://arxiv.org/abs/2510.13430</a> )                                                                                                                                                                                                      |

### Requires Expert Verification

- Which specific contested questions in the full dataset would require multi-perspective treatment under the deployment's pluralistic framing

## Remediation

**Gap 1: Single-correct-answer MCQ format cannot represent multi-perspective acknowledgment required by deployment [Q27, elicitation A3]**

**Recommendation:** Supplement ArabicMMLU with an open-ended Arabic generation evaluation (e.g., AraGen 3C3H [WEB-6] or 'Beyond MCQ' [WEB-5]) and develop a custom rubric that scores perspective-flagging on contested historical/civic items, with explicit credit for acknowledging country-specific framings.

| ID    | Evidence                                                                                                                                                        |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q27   | "Each question has 2–5 candidate answers, with one correct answer." (p.3)                                                                                       |
| WEB-5 | 'Beyond MCQ' Arabic open-ended benchmark — supplementary alternative for OF ( <a href="https://arxiv.org/abs/2510.24328">https://arxiv.org/abs/2510.24328</a> ) |
| WEB-6 | AraGen 3C3H LLM-as-judge framework for open-ended Arabic generation ( <a href="https://arxiv.org/abs/2510.13430">https://arxiv.org/abs/2510.13430</a> )         |

**Score: 2/5 — Significant gaps** | **Confidence: high**

**Benchmark:** ARABICMMLU | **Context:** Non-Arab Tourists and Expats in Eight Arabic-Speaking Countries (ArabicMMLU Deployment)

**Output Content definition:** Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

Label-correctness verification process — 96% inter-annotator agreement on 100 sampled questions [Q42, Q43] with native MSA speakers holding Bachelor's degrees and competitive compensation [Q35, Q36, Q39] — establishes a reasonable practical ceiling [Q44]. However, annotator-pool composition undermines convergent validity for the multi-country deployment: 10 annotators are from Jordan (4), Egypt (2), Lebanon (1), UAE (1), KSA (2), with no Moroccan, Kuwaiti, or Palestinian annotators [Q31] despite Palestine being a top-3 source country [WEB-3, WEB-4]. For contested historical questions where 'correct' answers vary across national curricula, the 96% agreement likely reflects Jordanian/Egyptian consensus rather than pan-regional validity. The sampled data shows concrete examples of Jordanian-curriculum framing for shared Arab history [DATASET-D1, DATASET-D2, DATASET-D3, DATASET-D11]. Training data contamination is acknowledged as unverifiable [Q95]. OCR corruption in university Egyptian items [DATASET-D23, DATASET-D24] suggests label quality may degrade in some sub-corpora. No aggregation method considerations for minority perspectives are documented.

| ID  | Evidence                                                                                                                                                                                                                                                                                                                                                          |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q31 | "The data construction process involved a total of 10 Arabic native speakers from different countries: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 from UAE, and 2 from KSA) and 4 external workers (3 Jordanian and 1 Egyptian)." (p.4)                                                                                                           |
| Q35 | "Our internal workers are Master's students and Research Assistants in Computer Science, while the external workers hold Bachelor's degrees." (p.4)                                                                                                                                                                                                               |
| Q36 | "We ensured competitive compensation for the workers, exceeding the monthly average wage in each respective country." (p.4)                                                                                                                                                                                                                                       |
| Q39 | "While our workers are native speakers of Modern Standard Arabic with at least Bachelor's degrees, we maintain the quality of our dataset construction through meticulous steps." (p.4)                                                                                                                                                                           |
| Q42 | "Finally, we assessed the accuracy of our data collection by having two native Arabic speakers annotate 100 randomly sampled questions. They were provided with all metadata, including the answer key, and tasked with verifying the correctness of each sample using any available resources (e.g., search engines)." (p.4)                                     |
| Q43 | "We found that 96% of the questions and answer keys match on average, while the remaining could have incorrect answer keys." (p.4)                                                                                                                                                                                                                                |
| Q44 | "This 96% score is considered to represent the maximum score meaningfully achievable for ArabicMMLU." (p.4)                                                                                                                                                                                                                                                       |
| Q95 | "to a lack of sufficient information about its training regimen. As such, we cannot assert that the model's pretraining data is free from contamination." (p.10)                                                                                                                                                                                                  |
| D1  | دحو مامت: ني طسلف يف اعامطأو ةي نو يوصل ة ك رحل نم لولأا هلل ادبع كل مل الم فق اوم نم تسيل ةي تال نم ة دحاو هضفر / ني طسلف ة مال قودنص مدع يل برعلا ةودع / قاربلا طئاح يل ارمم دوهيلا اعطاع هضفر / م 1950 ماع ني تفضل الم<br>Jordan HS history question about King Abdullah I's positions on Zionism and Palestine — framed from Jordanian curriculum perspective |
| D2  | نب لصل يف رم ٢٠١ / ني سحلا نب هلل ادبع رم ١٠٠: سي راب يف ءافلحل ال دقع يذل ا لصل رمتؤم يف برعلا _____ لثم<br>Jordan civics HS: who represented Arabs at Paris Peace Conference — Jordanian Hashemite framing                                                                                                                                                      |
| D3  | Jordan civics HS: — بادتالال كف دويقي نم دالبل ررح: ١٩٢٨ ماع هي نا طي ربل الهي ندرألا هدهاع ملل سي رئلل فدهل ناك<br>primary goal of the 1948 Jordanian-British treaty — Jordanian national framing                                                                                                                                                                |
| D11 | Jordan primary social science: fact about King Talal's reign — Jordan-specific<br>في كل ذو ةي حص باب سأل هت يالو تهتنا ني سحلا نب لال كل مل الم                                                                                                                                                                                                                   |
| D23 | Severely garbled OCR text making question almost illegible<br>ة زكرم لا: لي ري تذت ... صا خا لال ني تب طل تذل ا عتي يه ت جرد                                                                                                                                                                                                                                      |

|              |                                                                                                                                                                                                                  |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>D24</b>   | قـبـس ام لك؟ مل اـرـسـالـا رـلـوـدـل اـرـف مـكـاـر لـا رـف اـر رـفـاـوت بـرر رـتـل اـرـسـالـا طـوـرـرـشـل اـنـرـم — Egypt university political science with severe OCR corruption — content partially unreadable |
| <b>WEB-3</b> | Palestine confirmed top-3 source country — yet no Palestinian annotators in pool ( <a href="https://github.com/mbzuai-nlp/ArabicMMLU">https://github.com/mbzuai-nlp/ArabicMMLU</a> )                             |
| <b>WEB-4</b> | HuggingFace dataset card confirms Palestine major contributor ( <a href="https://huggingface.co/datasets/MBZUAI/ArabicMMLU">https://huggingface.co/datasets/MBZUAI/ArabicMMLU</a> )                              |

## Strengths

- 96% answer-key accuracy verification on a 100-question sample provides a documented quality ceiling [Q42, Q43, Q44]
- Native-speaker annotator workforce with at least Bachelor's degrees and competitive pay [Q35, Q36, Q39]
- Pre-collection workshop standardized annotator process [Q40]
- Annotators include workers from 5 of 8 deployment countries (Jordan, Egypt, Lebanon, UAE, KSA) [Q31, Q96]

| ID         | Evidence                                                                                                                                                                                                                                                                                                                      |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q31</b> | "The data construction process involved a total of 10 Arabic native speakers from different countries: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 from UAE, and 2 from KSA) and 4 external workers (3 Jordanian and 1 Egyptian)." (p.4)                                                                       |
| <b>Q35</b> | "Our internal workers are Master's students and Research Assistants in Computer Science, while the external workers hold Bachelor's degrees." (p.4)                                                                                                                                                                           |
| <b>Q36</b> | "We ensured competitive compensation for the workers, exceeding the monthly average wage in each respective country." (p.4)                                                                                                                                                                                                   |
| <b>Q39</b> | "While our workers are native speakers of Modern Standard Arabic with at least Bachelor's degrees, we maintain the quality of our dataset construction through meticulous steps." (p.4)                                                                                                                                       |
| <b>Q40</b> | "Firstly, we conducted a 1-hour workshop before the data collection stage to clarify the process." (p.4)                                                                                                                                                                                                                      |
| <b>Q42</b> | "Finally, we assessed the accuracy of our data collection by having two native Arabic speakers annotate 100 randomly sampled questions. They were provided with all metadata, including the answer key, and tasked with verifying the correctness of each sample using any available resources (e.g., search engines)." (p.4) |
| <b>Q43</b> | "We found that 96% of the questions and answer keys match on average, while the remaining could have incorrect answer keys." (p.4)                                                                                                                                                                                            |
| <b>Q44</b> | "This 96% score is considered to represent the maximum score meaningfully achievable for ArabicMMLU." (p.4)                                                                                                                                                                                                                   |
| <b>Q96</b> | "We extend our gratitude to all collaborators from Jordan, Egypt, Lebanon, UAE, and Saudi Arabia who participated in the data collection process." (p.10)                                                                                                                                                                     |

## Checklist

### Checklist item definitions:

| ID          | Assessment Question                                               |
|-------------|-------------------------------------------------------------------|
| <b>OC-1</b> | Do ground-truth labels reflect regional stakeholder perspectives? |
| <b>OC-2</b> | Assess potential annotator-regional population disagreement       |
| <b>OC-3</b> | Review annotator demographics from documentation                  |
| <b>OC-4</b> | Consider label re-annotation by regional annotators               |
| <b>OC-5</b> | Review aggregation methods for minority perspective erasure       |
| <b>OC-6</b> | Document label issues harming convergent/external validity        |

### Checklist responses:

**OC-1:** Partially — labels reflect curriculum-stated 'correct' answers from source exams, which encode national perspectives. For non-contested STEM content this is acceptable [DATASET-D21, DATASET-D38]; for contested civic/historical content [DATASET-D1, DATASET-D3], labels reflect source-country curriculum perspective without acknowledgment.

**OC-2:** High potential for disagreement on contested content — Palestinian, Moroccan, and Kuwaiti perspectives are not represented in the verifier pool [Q31] despite Palestine being a major source [WEB-3, WEB-4]. The 96% figure [Q43] likely overstates regional validity for politically sensitive items.

**OC-3:** Documented: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 UAE, 2 KSA), 4 external workers (3 Jordanian, 1 Egyptian) [Q31]; internal workers are Master's/RA-level CS students, external workers hold Bachelor's [Q35]. Demographics are documented but skew Jordanian/Egyptian.

**OC-4:** Re-annotation by Palestinian, Moroccan, and Kuwaiti annotators recommended for contested-content items, especially history/civics from those source countries.

**OC-5:** No aggregation method for minority/competing perspectives is documented — the benchmark's structure precludes it [Q27].

**OC-6:** Documented: (1) annotator-pool skew toward Jordan/Egypt with absent Moroccan/Palestinian/Kuwaiti perspectives [Q31, Q96]; (2) 96% ceiling does not account for cross-curriculum disagreement on contested content [Q43, Q44]; (3) OCR corruption in some sub-corpora [DATASET-D23, DATASET-D24]; (4) contamination unverifiable [Q95].

| ID    | Evidence                                                                                                                                                                                                                                                                                                                                                                                              |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q27   | "Each question has 2–5 candidate answers, with one correct answer." (p.3)                                                                                                                                                                                                                                                                                                                             |
| Q31   | "The data construction process involved a total of 10 Arabic native speakers from different countries: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 from UAE, and 2 from KSA) and 4 external workers (3 Jordanian and 1 Egyptian)." (p.4)                                                                                                                                               |
| Q35   | "Our internal workers are Master's students and Research Assistants in Computer Science, while the external workers hold Bachelor's degrees." (p.4)                                                                                                                                                                                                                                                   |
| Q43   | "We found that 96% of the questions and answer keys match on average, while the remaining could have incorrect answer keys." (p.4)                                                                                                                                                                                                                                                                    |
| Q44   | "This 96% score is considered to represent the maximum score meaningfully achievable for ArabicMMLU." (p.4)                                                                                                                                                                                                                                                                                           |
| Q95   | "to a lack of sufficient information about its training regimen. As such, we cannot assert that the model's pretraining data is free from contamination." (p.10)                                                                                                                                                                                                                                      |
| Q96   | "We extend our gratitude to all collaborators from Jordan, Egypt, Lebanon, UAE, and Saudi Arabia who participated in the data collection process." (p.10)                                                                                                                                                                                                                                             |
| D1    | دحو ومامت: ني طسلف يف اهعاطأو ةي نوي هصلال ةكرحل نم لوألا هلل ادبع كل مل فق او م نم تسيل ةي تآلا نم ةدحاو هضفر / ني طسلف ل ةمألا قودنص م غد ل برعلا ةودع / قاربلا طئاح ل ارم دوهيلا اعطع هضفر / م 1950 ماع ني تفضلال اعطاق اضر فر روفل دعوو وك ي سكي اس ةدهاعم — Jordan HS history question about King Abdullah I's positions on Zionism and Palestine — framed from Jordanian curriculum perspective |
| D3    | Jordan civics HS: بادت نال ك ف دوي ق نم دال بل رحت: ١٩٢٨ ماع هي ناطي ربل هي ن ذرألا هدهاعم لل سي ئرل فدهل ناك primary goal of the 1948 Jordanian-British treaty — Jordanian national framing                                                                                                                                                                                                          |
| D21   | تادي ببل ل عي نصت نع ةل وئسم ةي ل اتال تاي ضعل ا دح / X100: وه ي ئوصل رهج مل يف ةي تي زلا ةس دعل ري بكت رادقم ةي ل خل يف — Palestine HS biology: standard cellular biology — culturally neutral STEM                                                                                                                                                                                                  |
| D23   | Severely garbled OCR text making question almost illegible — ...صاختلألا ني تب ةطل تدل عتي ي هت ةت جرد                                                                                                                                                                                                                                                                                                |
| D24   | Egypt university political science with severe OCR corruption — content partially unreadable — ق ب س ام لك م ار رسال رل و دل ا رف م كر ار ل ا رف ار رفاوت ب رر رر ل ا رسال ا طور رر شل ن ررم                                                                                                                                                                                                          |
| D38   | + موي سي ن غاملا ل ص ح ي ذل ل عاف تل فصت ةي ل اتال تال داعملا ي ا رب تخم ل اب موي سي ن غاملا نم ةي م ك قرح ب دم ا ماق — Natural science MS: chemistry reactions — country-neutral STEM موي سي ن غاملا دي سك ا -> ني ج س ك ا                                                                                                                                                                           |
| WEB-3 | Palestine confirmed top-3 source country — yet no Palestinian annotators in pool ( <a href="https://github.com/mbzuai-nlp/ArabicMMLU">https://github.com/mbzuai-nlp/ArabicMMLU</a> )                                                                                                                                                                                                                  |
| WEB-4 | HuggingFace dataset card confirms Palestine major contributor ( <a href="https://huggingface.co/datasets/MBZUAI/ArabicMMLU">https://huggingface.co/datasets/MBZUAI/ArabicMMLU</a> )                                                                                                                                                                                                                   |

## Information Gaps

- Whether the 100-question verification sample included contested-content items
- Distribution of OCR corruption across the full dataset

## Requires Expert Verification

- Whether Palestinian, Moroccan, and Kuwaiti subject-matter experts would agree with ground-truth labels on country-sourced contested items

## Remediation

**Gap 1: Annotator pool lacks Palestinian, Moroccan, and Kuwaiti members despite Palestine being a top-3 source country [Q31, WEB-3]**

**Recommendation:** Re-verify a stratified sample of Palestinian-, Moroccan-, and Egyptian-sourced contested-content items with annotators from those countries; report disagreement rates separately from the 96% pooled figure to expose curriculum-specific disagreement.

| ID    | Evidence                                                                                                                                                                                                                                                |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q31   | "The data construction process involved a total of 10 Arabic native speakers from different countries: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 from UAE, and 2 from KSA) and 4 external workers (3 Jordanian and 1 Egyptian)." (p.4) |
| WEB-3 | Palestine confirmed top-3 source country — yet no Palestinian annotators in pool ( <a href="https://github.com/mbzuai-nlp/ArabicMMLU">https://github.com/mbzuai-nlp/ArabicMMLU</a> )                                                                    |

# Output Form (OF)

Score: 1/5 — Serious concern | Confidence: high

**Benchmark:** ARABICMMLU | **Context:** Non-Arab Tourists and Expats in Eight Arabic-Speaking Countries (ArabicMMLU Deployment)

**Output Form definition:** Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

The benchmark measures performance as percentage MCQ accuracy [Q3, Q45] via highest-token-probability scoring for open-source models [Q50, Q79] or first-token regex matching for closed-source models [Q53, Q54]. This output form is fundamentally misaligned with the deployment's requirement for open-ended, explanatory, multi-perspective natural-language responses [elicitation A3]. The benchmark cannot evaluate whether a model produces appropriately nuanced, perspective-flagging answers — it can only score whether the model selects the curriculum-designated correct option. Additional output-form analyses (calibration [Q80], length-confidence correlation [Q82], few-shot trends [Q75, Q76], negation sensitivity [Q83, Q84]) all operate on the same MCQ scoring surface. The text-based MCQ form is appropriate for literacy-capable populations [Q93] in principle, but the form mismatch with the deployment's natural-language explanatory output requirement is structural and not addressable through reweighting or filtering. Supplementary frameworks (AraGen 3C3H, 'Beyond MCQ') exist [WEB-5, WEB-6] but are separate evaluations.

| ID    | Evidence                                                                                                                                                                                                                              |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q3    | "Our comprehensive evaluations of 35 models reveal substantial room for improvement, particularly among the best open-source models." (p.1)                                                                                           |
| Q45   | "Our experiments focus on zero-shot and few-shot settings across 35 models." (p.5)                                                                                                                                                    |
| Q50   | "Following previous studies (Koto et al., 2023; Li et al., 2023), for open-source models, we determine the answer based on the highest probability among all possible options." (p.6)                                                 |
| Q53   | "For closed-source models, we determine the answer based on the first token generated in the text using a regular expression." (p.6)                                                                                                  |
| Q54   | "If there is no match, we assign a random answer." (p.6)                                                                                                                                                                              |
| Q75   | "While all the results in Section 4.2 were based on zero-shot learning, we observe in Figure 7 that when we move to few-shot learning, results for base models improve but those for instruction-tuned models deteriorate." (p.8)     |
| Q76   | "Specifically, AceGPT and Jais show an improvement of 2–10 points when using few-shot learning, but the results for BLOOMZ and Jais-chat drop." (p.8)                                                                                 |
| Q79   | "The answer probability is obtained through softmax normalization across the available candidate answers." (p.8)                                                                                                                      |
| Q80   | "In Figure 8, we observe that the three open-source models are well calibrated with correlation scores $r > 0.9$ ." (p.8)                                                                                                             |
| Q82   | "We find no correlation between the length of the questions and the model confidence for either Jais or AceGPT." (p.8)                                                                                                                |
| Q83   | "Despite negation being an absolutely foundational linguistic phenomenon, LLMs have been shown to be worryingly insensitive to its effects in English (Kassner and Schütze, 2020; Hosseini et al., 2021; Truong et al., 2023)." (p.8) |
| Q84   | "We thus perform an analysis of LLM performance over questions in ArabicMMLU with and without negation to determine whether this observation ports across to Arabic." (p.8)                                                           |
| Q93   | "ArabicMMLU is focused solely on text-based assessment, and the exploration of multimodal questions is left for future work." (p.9)                                                                                                   |
| WEB-5 | 'Beyond MCQ' Arabic open-ended benchmark — supplementary alternative for OF ( <a href="https://arxiv.org/abs/2510.24328">https://arxiv.org/abs/2510.24328</a> )                                                                       |
| WEB-6 | AraGen 3C3H LLM-as-judge framework for open-ended Arabic generation ( <a href="https://arxiv.org/abs/2510.13430">https://arxiv.org/abs/2510.13430</a> )                                                                               |

## Strengths

- Standardized accuracy metric is reproducible and comparable across 35 evaluated models [Q3, Q45]

- Auxiliary analyses (calibration, length-confidence, negation sensitivity) provide robustness signal beyond raw accuracy [Q80, Q82, Q83]
- Two scoring methods documented for open-source vs closed-source models [Q50, Q53]

| ID  | Evidence                                                                                                                                                                                                                              |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q3  | "Our comprehensive evaluations of 35 models reveal substantial room for improvement, particularly among the best open-source models." (p.1)                                                                                           |
| Q45 | "Our experiments focus on zero-shot and few-shot settings across 35 models." (p.5)                                                                                                                                                    |
| Q50 | "Following previous studies (Koto et al., 2023; Li et al., 2023), for open-source models, we determine the answer based on the highest probability among all possible options." (p.6)                                                 |
| Q53 | "For closed-source models, we determine the answer based on the first token generated in the text using a regular expression." (p.6)                                                                                                  |
| Q80 | "In Figure 8, we observe that the three open-source models are well calibrated with correlation scores $r > 0.9$ ." (p.8)                                                                                                             |
| Q82 | "We find no correlation between the length of the questions and the model confidence for either Jais or AceGPT." (p.8)                                                                                                                |
| Q83 | "Despite negation being an absolutely foundational linguistic phenomenon, LLMs have been shown to be worryingly insensitive to its effects in English (Kassner and Schütze, 2020; Hosseini et al., 2021; Truong et al., 2023)." (p.8) |

## Checklist

### Checklist item definitions:

| ID   | Assessment Question                                    |
|------|--------------------------------------------------------|
| OF-1 | Does output modality match regional deployment needs?  |
| OF-2 | Assess text-to-speech availability for speech output   |
| OF-3 | Consider literacy rates and accessibility requirements |
| OF-4 | Document form mismatches harming external validity     |

### Checklist responses:

**OF-1:** Mismatch — benchmark output is discrete MCQ labels [Q3, Q50, Q53]; deployment requires open-ended natural-language responses with multi-perspective acknowledgment [elicitation A3, output\_form\_mismatch].

**OF-2:** Not applicable — deployment is text-only [A4], so TTS availability is not required.

**OF-3:** Target population is literate non-native Arabic learners [target\_population\_description]; text output is appropriate, but the benchmark's discrete-label form does not assess natural-language fluency, explanatory quality, or perspective-flagging.

**OF-4:** Documented: external validity violation — benchmark MCQ accuracy will not predict whether the system produces appropriately explanatory, multi-perspective natural-language responses required by the deployment [elicitation A3, output\_form\_mismatch.mismatch\_severity=HIGH].

| ID  | Evidence                                                                                                                                                                              |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q3  | "Our comprehensive evaluations of 35 models reveal substantial room for improvement, particularly among the best open-source models." (p.1)                                           |
| Q50 | "Following previous studies (Koto et al., 2023; Li et al., 2023), for open-source models, we determine the answer based on the highest probability among all possible options." (p.6) |
| Q53 | "For closed-source models, we determine the answer based on the first token generated in the text using a regular expression." (p.6)                                                  |
| Q93 | "ArabicMMLU is focused solely on text-based assessment, and the exploration of multimodal questions is left for future work." (p.9)                                                   |
| Q94 | "It is important to emphasize that our experimental results do not provide conclusive answers regarding the performance of LLMs in Arabic." (p.9)                                     |

## Remediation

**Gap 1: MCQ accuracy scoring cannot evaluate explanatory natural-language quality required by deployment**  
**[Q3, Q50, Q53]**

**Recommendation:** Adopt LLM-as-judge or human-rater scoring of free-form responses to a curated subset of ArabicMMLU history/civics/Islamic-studies questions reformulated as open-ended questions, scoring on correctness, completeness, and multi-perspective coverage.

| ID  | Evidence                                                                                                                                                                              |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q3  | “Our comprehensive evaluations of 35 models reveal substantial room for improvement, particularly among the best open-source models.” (p.1)                                           |
| Q50 | “Following previous studies (Koto et al., 2023; Li et al., 2023), for open-source models, we determine the answer based on the highest probability among all possible options.” (p.6) |
| Q53 | “For closed-source models, we determine the answer based on the first token generated in the text using a regular expression.” (p.6)                                                  |